Angles on Parallel Lines

Name _____ Date _____ No calculator Version 1

Scan to practise this topic online

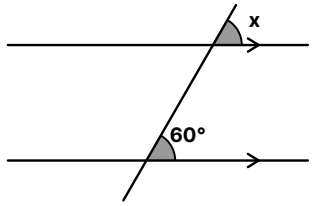
KEY WORDS parallel *(whakarara)* · transversal *(rārangi whakawhiti)* · corresponding *(koki hāngai-rite)*

A Corresponding angles ●○○

Remember: Corresponding angles are in the same position at each crossing. On parallel lines they are equal. Look for an F shape.

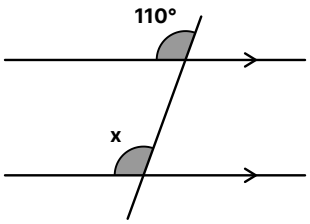
1. The arrows show that the lines are parallel. Find the size of the angle marked x.

a)



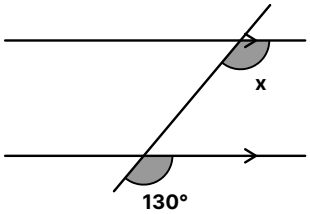
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b)



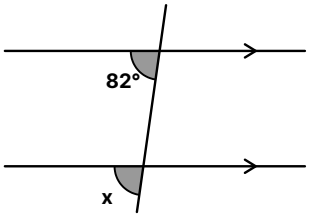
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c)



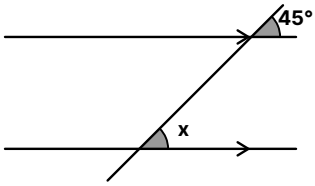
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d)



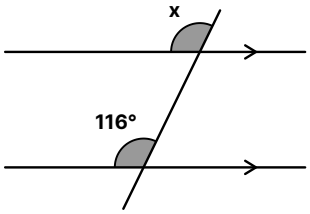
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e)



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f)



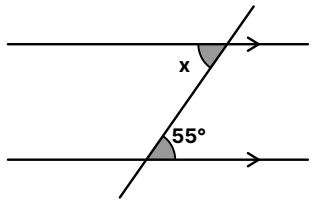
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B Alternate angles ●○○

Remember: Alternate angles are between the parallel lines, on opposite sides of the line that crosses them. They are equal. Look for a Z shape.

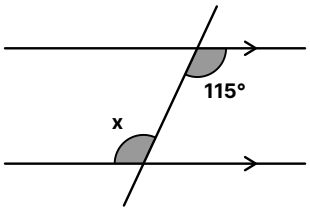
2. Find the size of the angle marked x.

a)



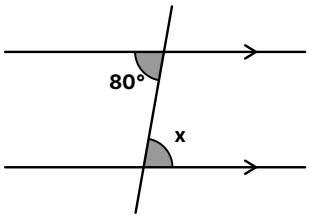
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b)



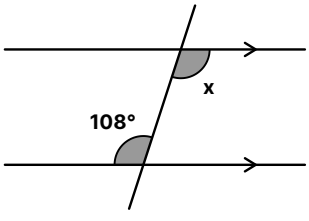
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c)



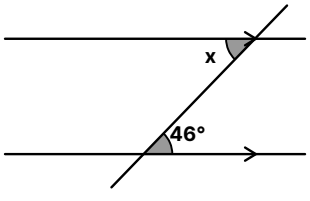
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d)



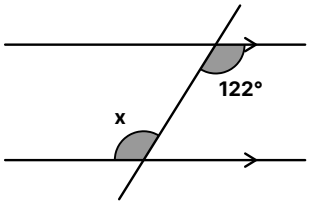
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e)



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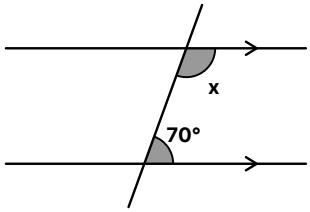
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C Co-interior angles ●●○

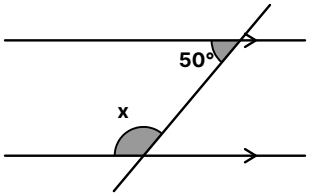
Remember: Co-interior angles are between the parallel lines, on the same side of the line that crosses them. They add to 180°. Look for a C shape.

3. Find the size of the angle marked x.

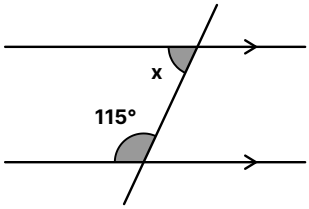
a)



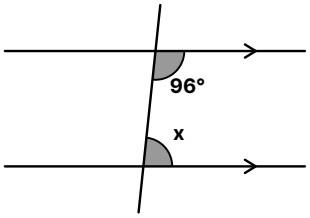
b)



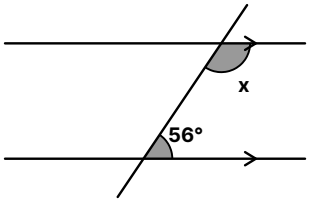
c)



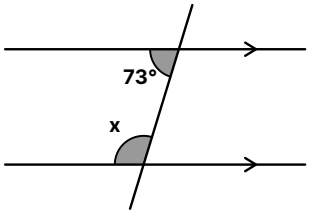
d)



e)



f)

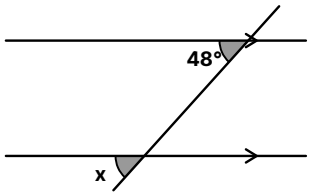


D Mixed, with reasons ●●●

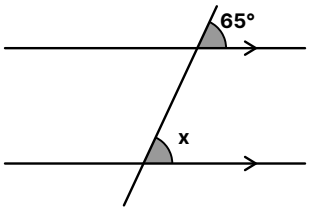
Watch out: Always give a reason. Say which rule you used: corresponding, alternate, co-interior, vertically opposite, or angles on a straight line.

4. Find the size of the angle marked x. Give a reason for each answer.

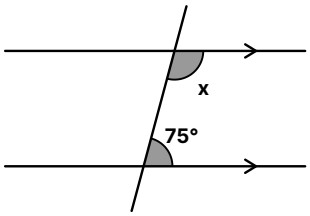
a)



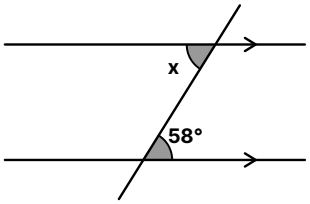
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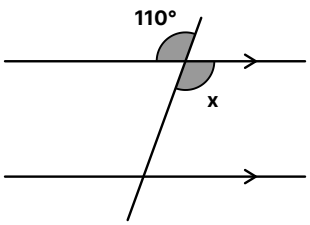
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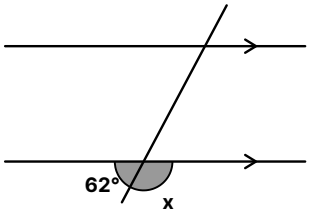
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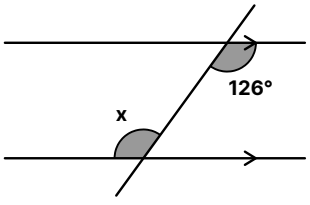
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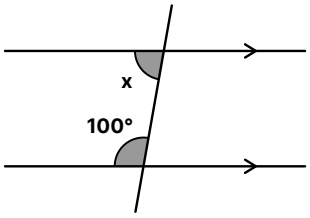
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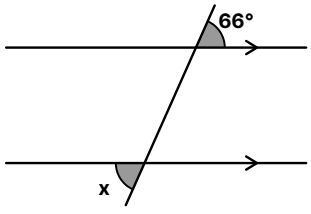
g)



h)



i)



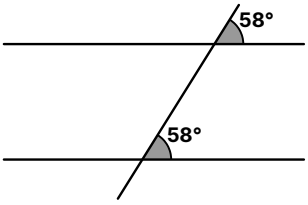
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E Are the lines parallel? ●●●

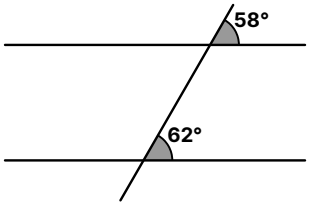
Remember: If a pair of corresponding or alternate angles are equal, or a pair of co-interior angles add to 180°, the lines are parallel.

5. These lines have no arrows. Decide whether the two lines are parallel, and give a reason.

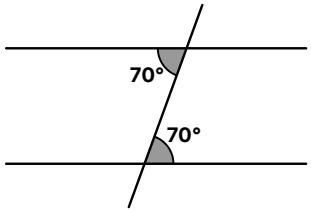
a)



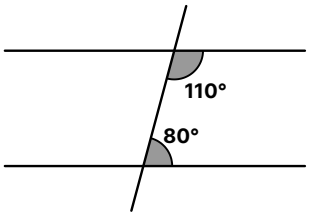
b)



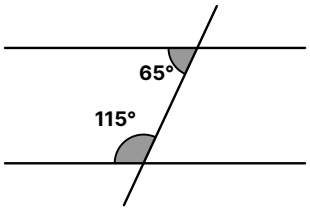
c)



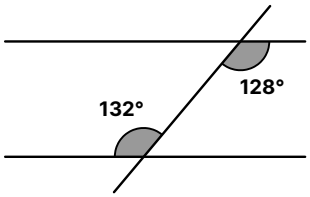
d)



e)



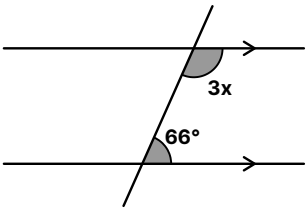
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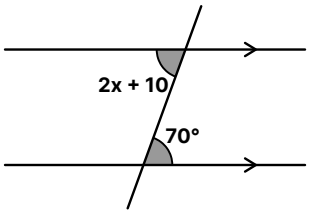
Extension

6. The lines are parallel. Write an equation and solve it to find x.

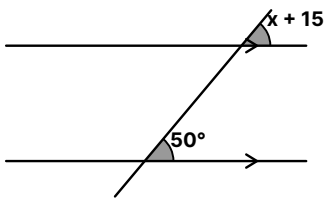
a)



b)



c)



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Angles on Parallel Lines

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

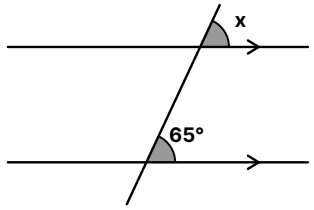
KEY WORDS parallel *(whakarara)* · transversal *(rārangi whakawhiti)* · corresponding *(koki hāngai-rite)*

A Corresponding angles ●○○

Remember: Corresponding angles are in the same position at each crossing. On parallel lines they are equal. Look for an F shape.

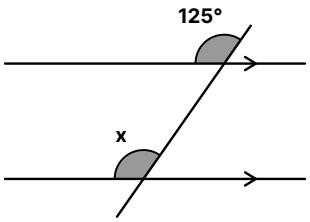
1. The arrows show that the lines are parallel. Find the size of the angle marked x.

a)



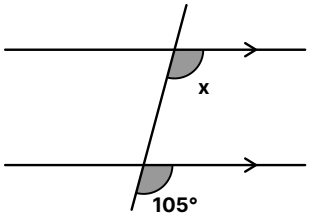
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b)



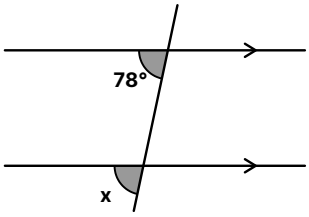
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c)



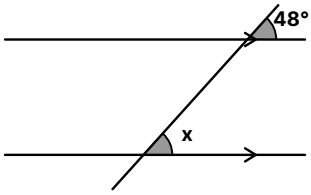
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d)



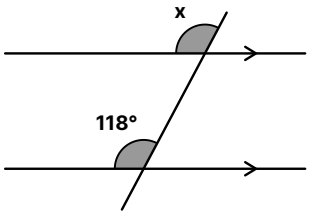
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e)



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f)



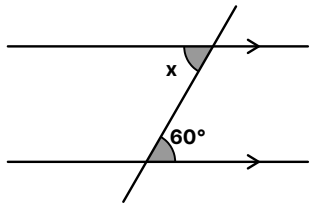
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B Alternate angles ●○○

Remember: Alternate angles are between the parallel lines, on opposite sides of the line that crosses them. They are equal. Look for a Z shape.

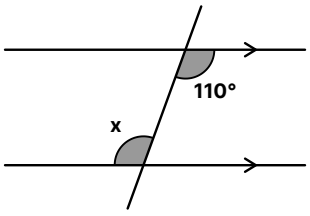
2. Find the size of the angle marked x.

a)



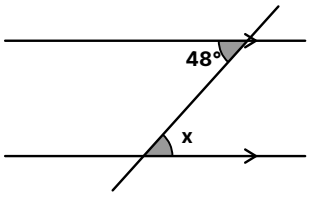
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b)



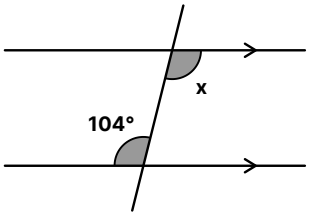
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c)



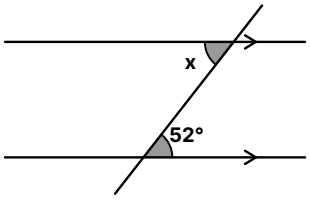
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d)



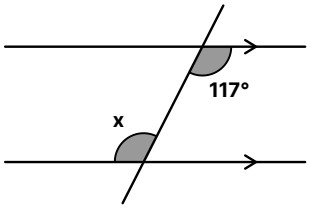
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e)



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f)



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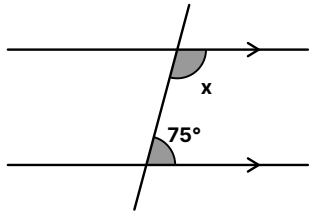
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C Co-interior angles ●●○

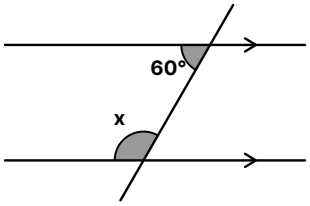
Remember: Co-interior angles are between the parallel lines, on the same side of the line that crosses them. They add to 180°. Look for a C shape.

3. Find the size of the angle marked x.

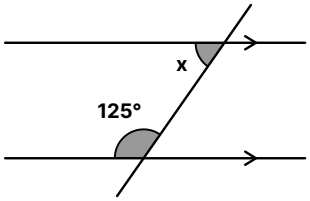
a)



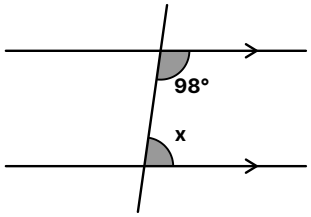
b)



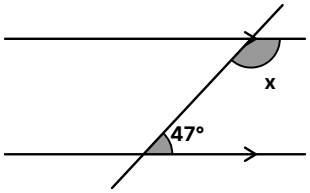
c)



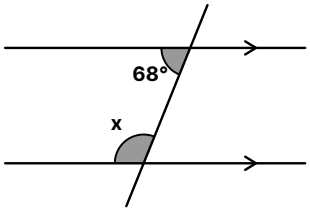
d)



e)



f)

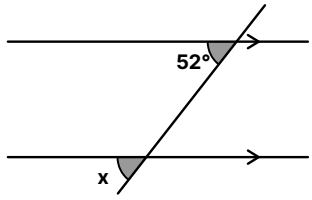


D Mixed, with reasons ●●●

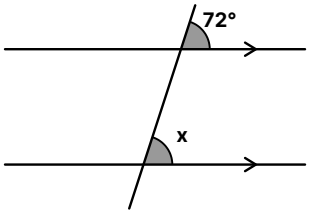
Watch out: Always give a reason. Say which rule you used: corresponding, alternate, co-interior, vertically opposite, or angles on a straight line.

4. Find the size of the angle marked x. Give a reason for each answer.

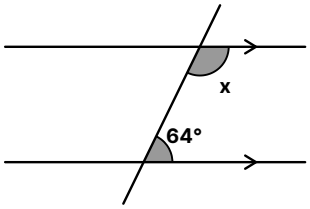
a)



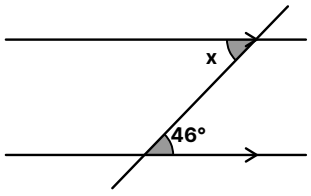
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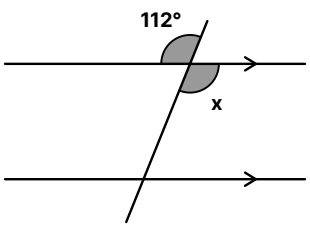
c)



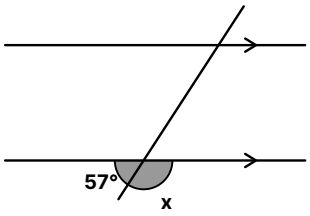
d)



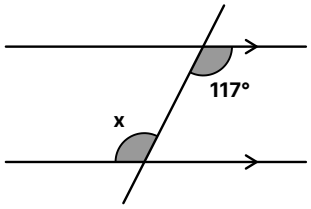
e)



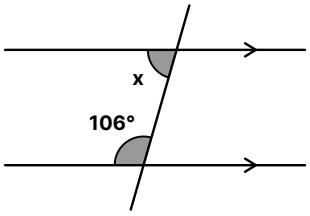
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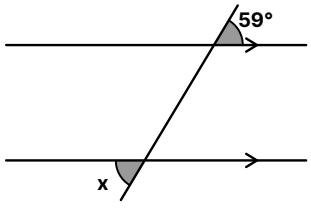
g)



h)



i)



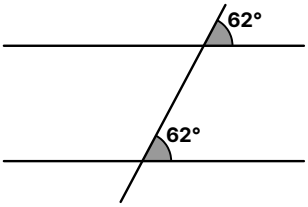
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E Are the lines parallel? ●●●

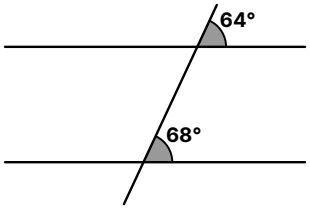
Remember: If a pair of corresponding or alternate angles are equal, or a pair of co-interior angles add to 180°, the lines are parallel.

5. These lines have no arrows. Decide whether the two lines are parallel, and give a reason.

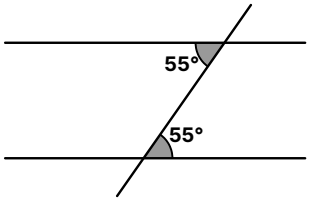
a)



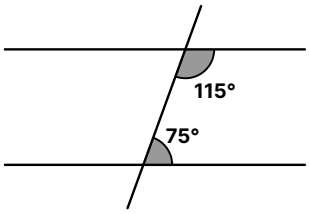
b)



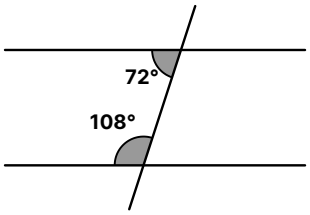
c)



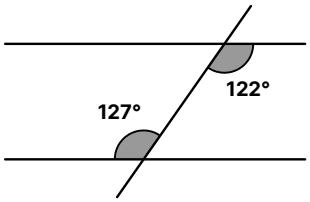
d)



e)



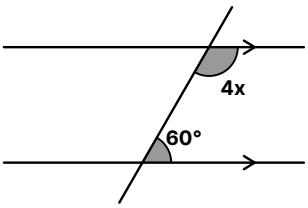
f)



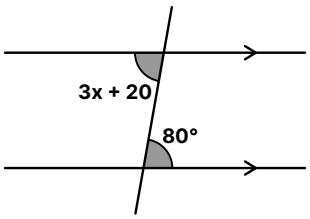
★ Extension

6. The lines are parallel. Write an equation and solve it to find x.

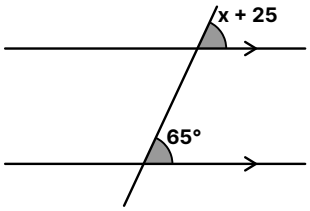
a)



b)



c)



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